Journal of Elder Abuse & Neglect

ISSN: 0894-6566 (Print) 1540-4129 (Online) Journal homepage: http://www.tandfonline.com/loi/wean20

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To cite this article: Brad Cannell, Jennifer M. Reingle Gonzalez, Melvin Livingston, Katelyn K. Jetelina, Jason Burnett & Julie C. Weitlauf (2019): Pilot testing the detection of elder abuse through emergency care technicians (DETECT) screening tool: results from the DETECT pilot project, Journal of Elder Abuse & Neglect

To link to this article: https://doi.org/10.1080/08946566.2018.1564104

Published online: 07 Jan 2019.

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JOURNAL OF ELDER ABUSE & NEGLECT
https://doi.org/10.1080/08946566.2018.1564104

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Pilot testing the detection of elder abuse through emergency care technicians (DETECT) screening tool: results from the DETECT pilot project

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ABSTRACT

Objective: To pilot test the feasibility of implementing an elder abuse (EA) screening tool (DETECT) designed for medics.

Methods: Testing occurred between September 17th, 2015 and October 26th, 2015. MedStar Mobile Healthcare medics completed the DETECT tool when responding to calls for community-dwelling patients 65 years of age or older.

Results: The DETECT tool was used 1,248 times by 97% of medics responding to an eligible 911 call. Medics responded affirmatively to at least one screening item on 209 of the completed screenings (16.8%). Immediately following the introduction of the DETECT tool, there was an increase of 5.4 (226% above baseline) reports per month ($p = 0.0056$).

Conclusions: The DETECT tool was easily incorporated into medic's field-based practice and resulted in an increase in medic generated reports of EA to APS. Future research designed to evaluate the tool's validity and reliability are warranted.

KEYWORDS
Elder Abuse; EMS; Screening

Introduction

The annual prevalence of elder abuse and neglect (EA) is about 11% for cognitively intact older adults, and estimated prevalence is even higher for those with cognitive impairment (Acierno et al., 2010; Connolly, Brandl, & Breckman, 2014). Many older adults experience chronic exposure to multiple forms of abuse including financial abuse/exploitation, neglect, and physical, emotional/psychological abuse, and/or sexual abuse (Connolly et al., 2014; National Center on Elder Abuse, 2017; National Research Council, 2003). Moreover, EA is linked with a host of consequences, including increased

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rates of hospital and nursing home admission, diminished physical and mental health, poor physical functioning, and heightened risk of mortality (Baker et al., 2009; Burnett et al., 2016; Cannell et al., 2015; Connolly et al., 2014; Mouton, Rodabough, Rovi, Brzyski, & Katerndahl, 2010; National Research Council, 2003). Nevertheless, identification or detection of EA is a complex undertaking, and there is broad consensus that many – perhaps the majority – of EA cases go undetected and unreported (Acierno et al., 2010; Connolly et al., 2014; Lifespan of Greater Rochester, Inc., Weill Cornell Medical Center of Cornell University, & New York City Department for the Aging, 2011; National Research Council, 2003).

Innovative research efforts that can help inform and guide effective and timely detection of EA are urgently needed to address this problem (National Institutes of Health, 2017). Emergency medical technicians and paramedics – collectively referred to as medics – are an important group of medical providers that can help in this effort. Medics are well positioned to identify cases of potential EA that would otherwise go undetected (Cannell, Jetelina, Zavadsky, & Gonzalez, 2016; Rosen et al., 2017). Indeed, older adults are four times more likely to use emergency medical services than younger adults (Shah et al., 2007). Further, medics are often granted unique access to the homes of older adults, which allows for the opportunity to observe the older adult's physical and social environment (Cannell et al., 2016). Such observations are typically not possible when older adults are seen in a clinic setting.

However, robust EA screening tools designed specifically to help medics detect EA in the field are not available. Currently available EA screening tools are lengthy and/or require the presence of a caregiver or direct questioning of the older adult (Fulmer & Cahill, 1984; Rathbone-McCuan, 1980; Ferguson & Beck, 1983; Fulmer, Guadagno, Bitondo Dyer, & Connolly, 2004; Straus, 1979; Hwalek, Sengstock, & Lawrence, 1984; Yaffe, Wolfson, Lithwick, & Weiss, 2008; Schofield & Mishra, 2003; Reis & Nahmiash, 1998b; Reis & Nahmiash, 1995; American Medical Association, 1993), rendering them inappropriate for use in an emergency response setting. For example, a tool that requires the medic to directly ask an older adult if they feel safe will often be impractical in an acute emergency situation (Cannell et al., 2016; Reingle Gonzalez, Cannell, Jetelina, & Radpour, 2016). However significant this quandary, no extant, evidence-based EA screening tools, provide an alternative.

The DETECT screening tool

In collaboration with Texas Adult Protective Services (APS) and one of the largest mobile healthcare providers in North Texas – MedStar Mobile Healthcare (MedStar) – our team developed and piloted an EA screening

tool: **D**etection of **E**lder Abuse **T**hrough **E**mergency **C**are **T**echnicians (DETECT). The DETECT tool was designed specifically to help medics identify *potential* EA among community-dwelling older adults during an emergency response. DETECT relies entirely on the medics' systematic observations of the older adults' physical and social environment – no direct questioning of the older adult or their caregivers is involved. The DETECT tool was developed through an iterative, user-centered design process in which input was gathered from key stakeholders, and revisions to the tool incorporated their feedback (Cannell et al., 2016; Reingle Gonzalez et al., 2016). The intent was for that process to result in an EA screening tool that was easy for medics to use in the field and that helped medics capture information about older adults, their environments, and their caregivers that is thought to be associated with the occurrence of EA (Cannell et al., 2016; Reingle Gonzalez et al., 2016).

Study aim

The long-term goal of the DETECT project is to refine the tool and fully validate its effectiveness in assisting first responders to link older adults who are potentially living with EA to needed programs and services. However, the goals of this preliminary 5-week pilot study were far more limited. Namely, our goals were to answer the following two questions about the feasibility of using the DETECT screening tool with medics in the field: (1) What procedural barriers, if any, might prevent DETECT from being used as intended by medics in the field?; and, (2) how often, if ever, are the characteristics/situations described in each of the DETECT screening items actually observed and recorded by medics in the field?

Methods

Study setting

We pilot tested the DETECT screening tool with medics in the field between September 17th and October 26th, 2015. During the pilot test, MedStar's Electronic Patient Care Reporting system (ePCR) was programmed to automatically prompt all medics to complete an EA screening using the DETECT tool while on an eligible 911 response. An eligible 911 response was defined as a call for a community-dwelling patient who was 65 years of age or older, the setting was the patient's residence, and the patient resided in the community (e.g., private home, unlicensed adult foster homes, unlicensed board and care homes, etc.). Other types of residences (e.g., licensed skilled nursing facilities) were excluded because reports of EA in these settings are generally not investigated by APS in Texas (Texas Department of Family and

Protective Services, 2018). By definition, older adults who participated in the pilot study had to live in MedStar's service area of an estimated 978,000 residents, which included Fort Worth, Texas, and 14 surrounding communities (MedStar Service Area/Member Cities, 2018).

Medic screening and reporting procedures

All medics employed by MedStar (N ≈ 300) were required to view a 6.5-minute online training video designed to inform them of the availability of the DETECT screening tool in their ePCR and provide them instructions for completing the tool. The training did not include any general guidance regarding warning signs that EA may be occurring to avoid introducing surveillance effects.

Figure 1 illustrates the steps involved in the screening and reporting procedures used by MedStar medics during the study. First, Medstar dispatched medics to 911 calls for emergency assistance per their usual standard operating procedures. If the 911 response was eligible, then the medic was prompted to use the DETECT screening tool.

If a medic checked "yes" to one or more screening items it was considered sufficient minimum evidence for a positive screen. Medics were instructed to report all positive screenings to APS per the current state mandate in Texas. It is important to note that the DETECT screening tool was not created to definitively determine if EA is occurring or not. Rather, it was designed to

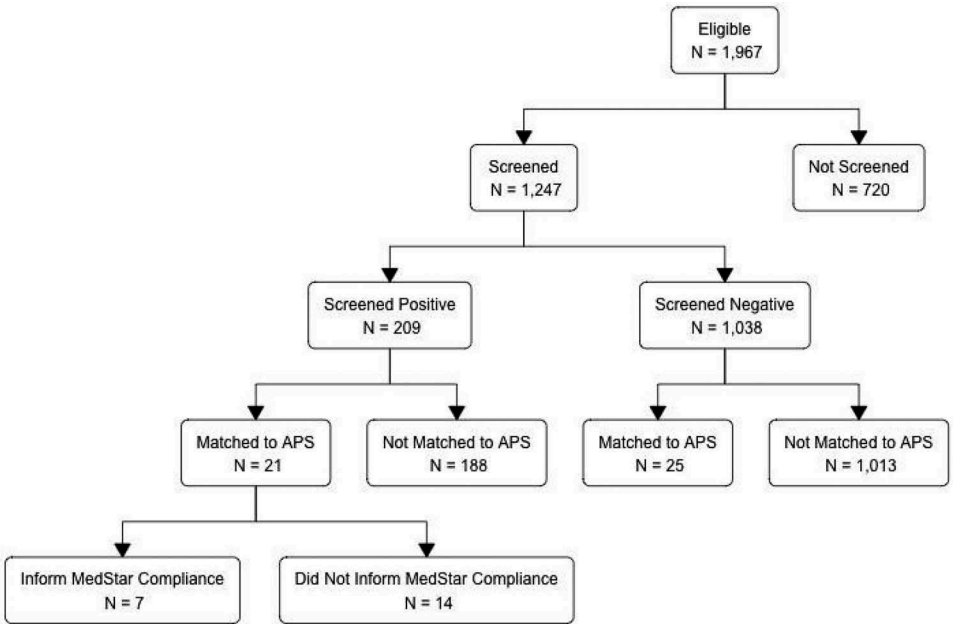

Figure 1. A flow diagram illustrating the DETECT pilot study screening and reporting procedures.

help medics determine if they should contact APS for further investigation and assistance. Therefore, a positive screen should not be interpreted as a confirmed instance of EA.

Finally, medics that contacted APS to report potential EA were instructed to make MedStar's internal legal compliance department aware of the APS report. This was the standard operating procedure that was in place at MedStar prior to the initiation of the DETECT pilot study. The feasibility of these screening and reporting procedures are evaluated below.

A push for universal adoption of DETECT during eligible calls

Initially, MedStar chose to implement the DETECT screening questions as an *optional* module in their ePCR. However, 11 days after beginning the pilot study (September 28th, 2015) MedStar decided to recommend universal application of the DETECT tool during eligible 911 calls. Specifically, systemic changes to the ePCR functionality were made such that it became difficult – but not impossible – to close-out an eligible patient's chart in the ePCR without also completing the DETECT screening tool. For context, in the ePCR multiple important modules are assigned "validation rules" (i.e., criteria for counting a module as completed). Satisfying each validation rule is associated with a certain number of "points" and a minimum number of "points" are required to close-out/submit a patient's chart. Therefore, after the validation rule was added to the DETECT screening module, it became more difficult for medics to complete a patient's chart without using the DETECT screening tool.

Data sources and measurement

We combined data from three different sources to investigate our study aims (Figure 2). These sources included:

(1) *MedStar ePCR records*. MedStar provided data on all eligible 911 responses during the study period. These data also included responses to the DETECT screening tool items in cases where the tool was used. These data were used to investigate the frequency and patterns of DETECT screening tool use in the field.
(2) *Texas APS investigations records*. We received data about all APS investigations conducted in MedStar's area of operations during the same time period that the DETECT pilot test took place. We then linked these records with the MedStar ePCR records (D1 above) using the patient's name, date of birth, and address. The combined MedStar/APS data was used to investigate the relationship between a positive DETECT screen and the occurrence of an APS investigation.

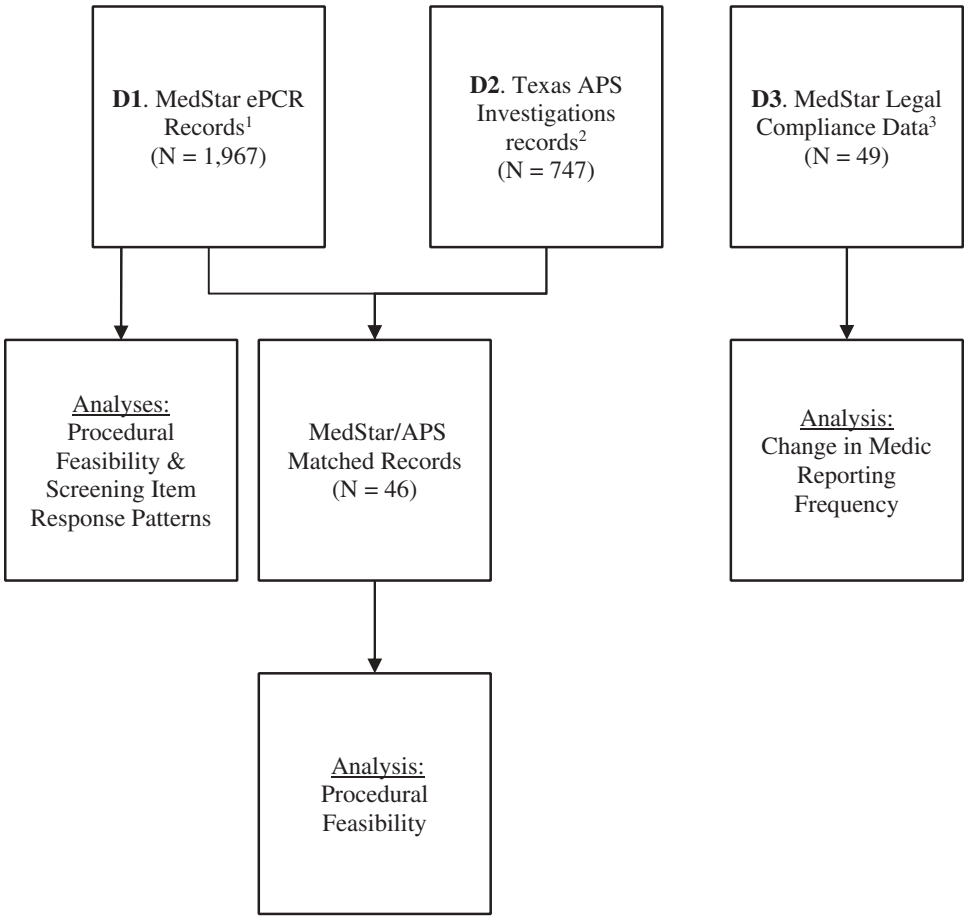

Figure 2. A flow diagram illustrating which data sources were used for each analysis. 1. Includes all uses of the DETECT screening tool (1,247).2. All APS investigations conducted in MedStar's area of operations during the same time period as the DETECT pilot test.3. All medic reports to APS in 2015 that MedStar's legal compliance department was made aware of.

(3) *MedStar legal compliance records*. MedStar's medical (ePCR) records are stored separately from their legal compliance department's records. Therefore, MedStar's legal compliance department provided all records of known APS reports made by their medics in 2015. These data were used to conduct a preliminary evaluation of the screening tool's impact on changes in medic reporting behavior.

Analytic methods

Procedural feasibility

To evaluate the uptake and use of the DETECT screening tool, we calculated the proportion of eligible older adults screened and the proportion of medics

that used the DETECT screening tool before and after the addition of the validation rule. Additionally, we calculated the number of DETECT screening tool items completed by medics on all eligible responses that took place during the study period. We also calculated the number and proportion of positive DETECT screenings with a matching record in the APS investigations data (D2) and in MedStar's legal compliance records (D3).

To examine the effect of the screening tool's introduction on the frequency of EA reporting in accordance with study procedures, we estimated the change in the mean of monthly EA reports to MedStar's legal compliance department before and after the introduction of DETECT. To account for non-independence due to repeated measures, we used robust standard errors in SAS v9.4's PROC GENMOD.

Screening item response patterns

The 26 DETECT screening items spanned 8 domains of EA indicators frequently encountered by medics (Cannell et al., 2016). The wording of each screening item is available in Table 1. For each item, response options included "Yes", "No", and "Don't know". Items related to the older adult's caregiver featured an expanded range of response options that included "Not Applicable or No Caregiver." For each of the screening items, we calculated the number and percent of times each possible response was observed (i.e, "Yes", "No", "Don't know", missing). We also calculated the count and percent of "Yes" responses that had a matching investigation record in the data we received from APS (D2 above). Finally, we calculated the number and percent of times that each item was the only item that received a "Yes" response.

The Committee for the Protection of Human Subjects at the University of Texas Health Science Center approved this project. Complete information about the definitions of terms used in APS investigations, analytic methods, additional results, and statistical programming code are available at https://github.com/brad-cannell/detect_pilot_test_5w.

Results

Procedural feasibility

The proportion of eligible older adults screened

During the pilot test, MedStar medics responded to 1,967 911 calls for adults aged 65 and older at their residence. Those 1,967 responses resulted in 1,247 DETECT screenings (63%). In the 11 days prior to the validation rule implementation, MedStar medics responded to 549 eligible 911 responses and completed 25 (5%) DETECT screenings. In the time period after implementing the validation rule, MedStar medics responded to 1,418 eligible 911 responses and completed 1,222 (86%) DETECT screenings. (Figure 3).

Table 1. Response patterns for each of the 26 DETECT screening item.

Screening Item	Completed[1] N (%)	Yes N (%)[2]	No N (%)[2]	Don't Know N (%)[2]	Yes – Only Yes N (%)[3]
Unusual odor (e.g. urine, feces)	1,246 (99.9%)	81 (6.5%)	1,117 (89.6%)	48 (3.9%)	37 (45.7%)
One or more utilities are not operational (no water electric, heating/cooling)	1,147 (92.0%)	6 (0.5%)	1,029 (89.7%)	112 (9.8%)	3 (50.0%)
Inside of the home is in extreme disarray/hoarding	1,150 (92.2%)	42 (3.7%)	1,044 (90.8%)	64 (5.6%)	8 (19.0%)
Living environment poses a health or safety concern (e.g. fire hazard, insect or rodent infestation, urine or feces present)	1,148 (92.1%)	42 (3.7%)	1,034 (90.1%)	72 (6.3%)	5 (11.9%)
Open alcohol containers or other alcohol/drug paraphernalia present	1,063 (85.2%)	9 (0.8%)	995 (93.6%)	59 (5.6%)	3 (33.3%)
If caregiver(s) present, they appear to lack knowledge of the patient/older adult's medical need(s)	681 (54.6%)	22 (3.2%)	563 (82.7%)	96 (14.1%)	2 (9.1%)
If caregiver(s) present, they appear unengaged and inattentive in caring for the patient/older adult	677 (54.3%)	11 (1.6%)	571 (84.3%)	95 (14.0%)	2 (18.2%)
If caregiver(s) present, they appear overwhelmed by the patient/older adult	677 (54.3%)	19 (2.8%)	561 (82.9%)	97 (14.3%)	4 (21.1%)
If caregiver(s) present, they appear frustrated, tired, angry, or burdened by the patient/older adult	677 (54.3%)	11 (1.6%)	568 (83.9%)	98 (14.5%)	0 (0.0%)
If caregiver(s) present, they appear overly concerned (e.g. anxious, hovering)	677 (54.3%)	10 (1.5%)	573 (84.6%)	94 (13.9%)	1 (10.0%)
If caregiver(s) present, they appear to be deceptive or withhold information from you	677 (54.3%)	3 (0.4%)	579 (85.5%)	95 (14.0%)	0 (0.0%)
If caregiver(s) present, they appear to give you information that is inconsistent with the patient/ older adult's account	677 (54.3%)	4 (0.6%)	578 (85.4%)	95 (14.0%)	1 (25.0%)
If caregiver(s) present, they appear to be under the influence of alcohol and/or illicit drugs	677 (54.3%)	3 (0.4%)	582 (86.0%)	92 (13.6%)	0 (0.0%)
If caregiver(s) present, they appear to be dependent on the patient/older adult for financial or other needs	677 (54.3%)	2 (0.3%)	572 (84.5%)	103 (15.2%)	0 (0.0%)
If caregiver(s) present, does the patient/older adult appear to lack social support or family that cares for them	674 (54.0%)	5 (0.7%)	569 (84.4%)	100 (14.8%)	0 (0.0%)
If caregiver(s) present, does the patient/older adult hesitate to talk with you or discuss their condition in front of the caregiver	674 (54.0%)	2 (0.3%)	566 (84.0%)	42 (6.2%)	2 (100.0%)
Is/Does the patient/older adult isolated in the home	710 (56.9%)	25 (3.5%)	641 (90.3%)	44 (6.2%)	9 (36.0%)
Is/Does the patient/older adult have suspicious falls, wounds, and/or injuries	710 (56.9%)	8 (1.1%)	687 (96.8%)	15 (2.1%)	2 (25.0%)
Is/Does the patient/older adult show evidence of any old injuries that have been left untreated	709 (56.9%)	6 (0.8%)	690 (97.3%)	13 (1.8%)	0 (0.0%)
Is/Does the patient/older adult under the influence of alcohol and/or illicit drugs	709 (56.9%)	13 (1.8%)	677 (95.5%)	19 (2.7%)	5 (38.5%)
Is/Does the patient/older adult appear depressed, anxious, or emotionally distressed for reasons other than their immediate medical condition	709 (56.9%)	12 (1.7%)	678 (95.6%)	19 (2.7%)	3 (25.0%)
Is/Does the patient/older adult have poor personal hygiene (including soiled in urine or feces)	708 (56.8%)	36 (5.1%)	660 (93.2%)	12 (1.7%)	2 (5.6%)
Is/Does the patient/older adult inadequately clothed or wearing dirty, torn, or soiled clothing	709 (56.9%)	27 (3.8%)	670 (94.5%)	12 (1.7%)	1 (3.7%)

(*Continued*)

Table 1. (Continued).

Screening Item	Completed[1] N (%)	Yes N (%)[2]	No N (%)[2]	Don't Know N (%)[2]	Yes – Only Yes N (%)[3]
Is/Does the patient/older adult have difficulties taking their prescribed medications as directed	709 (56.9%)	20 (2.8%)	640 (90.3%)	49 (6.9%)	1 (5.0%)
Is/Does the patient/older adult appear to be hoarding/ saving old medications	708 (56.8%)	6 (0.8%)	656 (92.7%)	46 (6.5%)	0 (0.0%)
Is/Does the patient/older adult need assistance with eating, toileting, transferring, dressing, or bathing	703 (56.4%)	61 (8.7%)	587 (83.5%)	55 (7.8%)	23 (37.7%)

[1] Count and percent of times each screening item was used (i.e., yes, no, or don't know). The denominator is the total number of DETECT screenings completed – 1,247.
[2] The denominator for this percentage is the number of times the screening item was used (column 2).
[3] When "yes" selected, the count and percentage of the time that it was it the only "yes" selected. The denominator is the number of "yes" selections (column 3).

The proportion of medics who completed a screening

During the pilot period, 258 medics participated in an eligible 911 response. Of those 258 medics, 251 (97%) completed at least one DETECT screening. Prior to the validation rule, 19% of medics who participated in an eligible 911 response completed at least one DETECT screening, compared to 99% after the validation rule.

The proportion of screening items completed

The distribution of screening items completed was bimodal, and not effectively characterized by the median. At 42% of DETECT screenings, medics

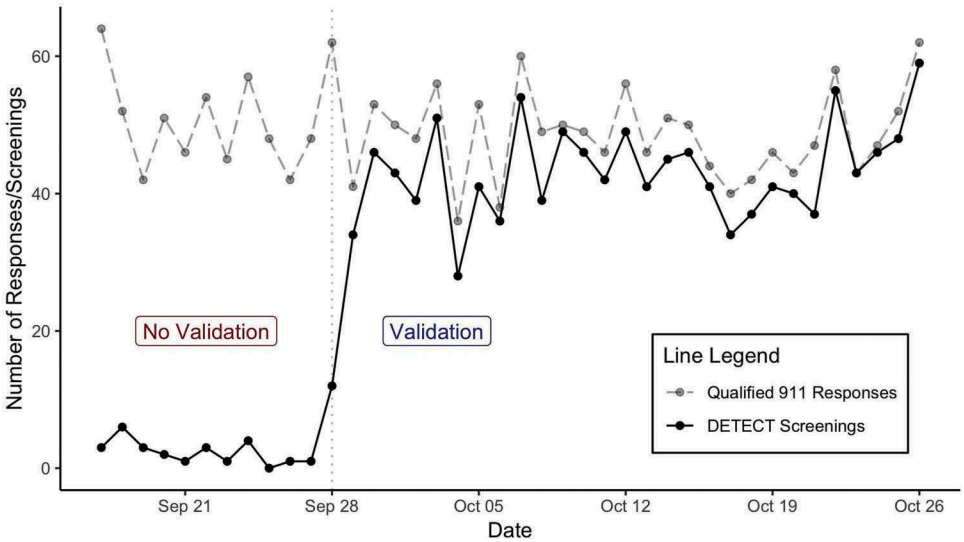

Figure 3. Qualified responses and completed screenings over the duration of the DETECT pilot study.

completed 5 or fewer screening items. Conversely, at 53% of the DETECT screenings, medics completed all 26 screening items.

Fidelity to the screening and reporting procedures

Figure 1 illustrates the steps involved in the screening and reporting procedures used by MedStar medics during the study. Of the 1,247 DETECT screenings completed, 209 (16.8%) were positive (i.e., a medic selected a "Yes" response to at least one screening item). We were able to match 21 of the 209 positive DETECT screenings (10%) to a completed APS investigation; however, we were also able to match 25 of the 1,038 negative DETECT screenings (2%) to a completed APS investigation. Finally, MedStar's legal compliance data indicated that 14 reports of EA were made to APS; however, only 7 of those reports could be matched to an APS investigation.

Change in EA reporting frequency

In 2015, MedStar's legal compliance department recorded 49 total reports to APS (Figure 4). During the five-week pilot study, 14 recorded reports to APS were documented by MedStar's legal compliance office. Results of our analysis indicated that immediately following the introduction of the DETECT screening tool, there was an increase of 5.4 ($p = 0.0056$) APS reports per month reported to MedStar's legal compliance department – a 226% improvement (Figure 4).

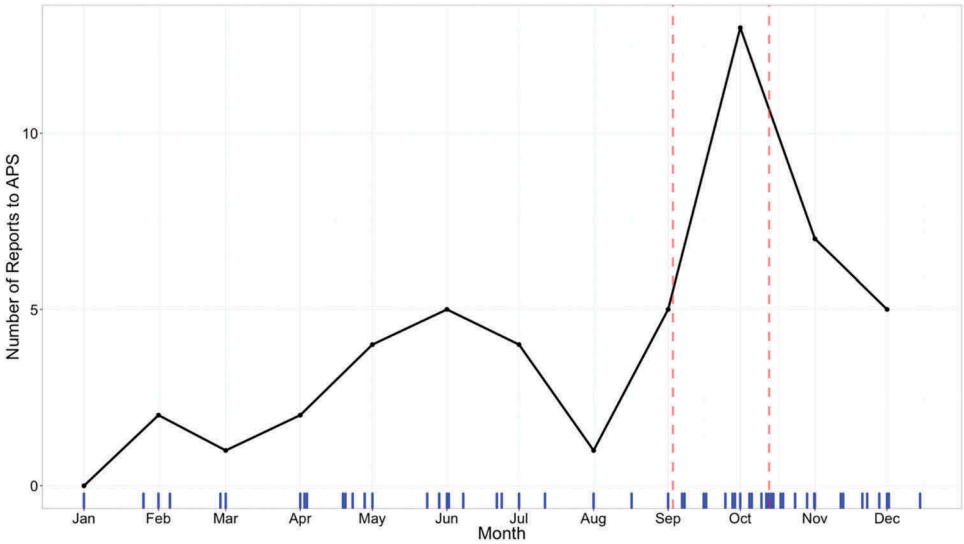

Figure 4. Medic reports to APS that MedStar's legal compliance department was made aware of, by month, 2015 (n = 49). 1. The black line represents total monthly reports.2. The red dashed lines indicate the time period in which the DETECT screening tool was used in the EPCR.3. The blue rug plot shows the marginal distribution of individual reports to APS.

Screening item response patterns

Complete details about the response patterns for each individual screening item are given in Table 1. The most frequently completed item (99.9% of screenings) was item number 1 – "Unusual odor (e.g. urine, feces)" and the most frequently skipped items were items 15 and 16 – "If caregiver(s) present, does the patient/older adult appear to lack social support or family that cares for them" and, "If caregiver(s) present, does the patient/older adult hesitate to talk with you or discuss their condition in front of the caregiver?", respectively. Both were completed at 54% of eligible screenings. "Yes" responses were selected for the individual screening items between 2 and 81 times. The item with the highest percentage of "Yes" responses (8.7%) was item number 26 – "Is/Does the patient/older adult need assistance with eating, toileting, transferring, dressing, or bathing?" The items with the lowest percentage of "Yes" responses (0.3%) were items number 14 and 16 – "If caregiver(s) present, they appear to be dependent on the patient/older adult for financial or other needs" and "If caregiver(s) present, does the patient/older adult hesitate to talk with you or discuss their condition in front of the caregiver?"

Discussion

The DETECT screening tool is the only screening tool designed specifically to be used by medics responding to emergency calls in the field. Results from this study provide preliminary evidence of the feasibility and utility of implementing the DETECT screening tool with medics to enhance the detection of EA. MedStar integrated the DETECT screening questions into its ePCR and pilot-tested the DETECT screening tool for over a month in the field. During that time, MedStar medics screened 1,247 older adults for EA in their homes. Further, the screening tool was used by 97% of the medics who responded to an eligible 911 call, and the number of APS reports per month reported to MedStar's legal compliance department increased by 5.4 reports per month (226%) post-implementation of the DETECT screening tool. Finally, each of the 26 screening items was observed multiple times by medics practicing in the field. These are all positive indicators that support the feasibility of using the DETECT screening tool in practice.

However, our results also highlight at least three indicators that the DETECT screening process was frequently not carried out as intended. First, during the study, MedStar added a validation rule to the DETECT module. Before the validation rule was put in place DETECT screenings were completed at only 5% of eligible 911 calls. After the establishment of the validation rule, DETECT screenings were completed in 86% of the eligible calls – soundly underscoring both the limitations of a "recommendation to

use" or "voluntary use" approach that relies on the clinician choosing to adopt a new screening strategy and the benefits of systemic adaptation of the clinical charting environment to promote consistent and appropriate use of a new screening tool. It also highlights the value of concurrent implementation and validation efforts with the introduction of a new screening tool in clinical practice. Although our current data do not allow us to comment on precisely why medics were less likely to use the screening tool without the validation rule, our previous qualitative work points to some potential explanations. For instance, our previous study found that medics reported time pressures during a 911 response (Reingle Gonzalez et al., 2016). These time pressures, together with the urgent nature of the medic work environment, can impact prioritization of activities – particularly those like completing a screening instrument – which may feel optional during an emergency medical response situation (Reingle Gonzalez et al., 2016).

Second, while all 26 items were completed at 53% of the screenings, 5 or fewer items were completed at 42% of the screenings, which may indicate that a much shorter screening tool would be ideal for maximizing completion rates. The notion that the ideal screening tool should be very brief is also supported through our prior qualitative research (Cannell et al., 2016). We did not have sufficient power to quantitatively identify redundant items in this pilot study. However, we did employ an expert panel of medics, emergency medicine administrators, and researchers to review the DETECT screening questions for redundancy. An observational study is currently underway to further test a shortened version of the screening tool. In this upcoming study, we will examine whether medics completed the shortened tool more frequently than the 26-item tool.

Third, the DETECT screening tool identified 209 older adults as being appropriate cases for referral to APS for further investigation (positive screen); yet, we were only able to match 21 (10%) of those cases with an APS investigation record. Further, only 7 (33%) of the cases that screened positive and were investigated by APS were reported to MedStar's legal compliance department in accordance with the study protocol and MedStar's standard operating procedures. It is possible that some or all of the 188 positive screens that were not reported to APS for further investigation were not reported because the medics conducting the screenings correctly believed that there was insufficient reason to suspect that the case was appropriate for an APS referral – regardless of receiving instructions to the contrary. In other words, the DETECT screening tool produces a high rate of false positives. Alternatively, it is possible that some or all of the 188 positive screens should have been reported to APS, but were not due to barriers to reporting that are unrelated to the validity of the tool itself. Indeed, in our prior qualitative work with medics we identified 5 common barriers to reporting including: (1) EMT apprehension towards violating older adults'

personal freedom to determine the conditions of their living environment; (2) EMT moral anxiety about the perceived negative consequences of an APS investigation on the older adult and/or their caregiver(s); (3) the time burden associated with reporting elder abuse or neglect to APS; (4) a perceived lack of case recall ability by EMTs; and, (5) low EMT confidence regarding ability to correctly identify potential elder abuse or neglect (Reingle Gonzalez et al., 2016). Therefore, further research is needed to fully investigate the cause of the discrepancy between the number of positive DETECT screenings and the number of APS investigations. Our team is currently planning two follow-up studies that aim to: (1) assess the DETECT tool's ability to correctly identify older adults who should be referred to APS for further investigation (i.e., validity) – independent of medic reporting practices, and (2) understand medics subjective beliefs about the value of the DETECT screening tool and what barriers to reporting, if any, it does/does not ameliorate (i.e., implementation).

Limitations

First, we understand that many readers will be tempted to extrapolate conclusions about the DETECT tool's validity, reliability and/or effectiveness from this study; however, we strongly caution against these inferences. Reliability, validity, and effectiveness are important aspects of the tool that will be fully assessed in the future, but were beyond the scope of this preliminary pilot study. Further research is needed to more comprehensively examine the concordance between DETECT screening results and the true occurrence of EA. For example, we were unable to compare DETECT screening results to APS investigation outcomes for 188 (90%) of the positive screens and 1,013 (98%) of the negative screens. As such, we were not able to calculate any measure of diagnostic performance that requires comparison to a gold standard measure (e.g., sensitivity and specificity).

Second, this pilot study was not designed to gather contextual information about EA cases, eclipsing the opportunity to fully understand the social, psychological, health, behavioral and environmental risk factors that contribute to EA. If these factors are observable in the older adult's environment, they may serve as early warning signs of EA.

Third, our study was limited to medics employed by MedStar who were providing care to older adults in the North Texas region. As such, the generalizability of our findings may be limited and caution should be exercised when extending the "lessons learned" from this study to other regions and populations.

To address these limitations, our future research will replicate these findings over a longer time frame using a gold standard validation mechanism

(e.g., an expert panel), and will include a validation rule that encourages medics to use the DETECT items in the ePCR across the entire duration of the study period. Upon completion of our developmental testing and refinement, our results should be replicated in other locations.

Despite these limitations, this study has many strengths. It is the first study to develop, integrate, and test the utility of an EA screening tool for use by medics. Our results suggest that the DETECT screening tool captures information on the occurrence of problematic environmental cues that may be associated with the occurrence of EA. Further, this study demonstrates the potential for the DETECT tool to increase the EA reporting rate – albeit with significant room for further improvement. With further validation, this tool may help address the widespread under-reporting of EA nationwide (Acierno et al., 2010; Lifespan of Greater Rochester, Inc. et al., 2011). Finally, the DETECT screening tool is based on medics' observations rather than questions; therefore, the tool has the potential to identify elder abuse or neglect regardless of injury, illness severity, culture or language.

Conclusion

In summary, medics commonly encounter probable EA exposed, community-dwelling older adults during responses to calls for medical services. The DETECT tool is the first field robust screening tool that enables medics to screen elders for EA exposure via observation – instead of interview – in the context of a broader emergency response visit. Our study provides preliminary evidence that the DETECT screening tool may be feasibly used in practice. Further, use of the DETECT tool appears to have substantially increased systematic screening for EA by medics and increased reporting of suspected EA cases among MedStar medics.

This work lays a solid empirical foundation regarding the feasibility of medics' implementation of a systematic observational EA screening tool, DETECT, in the field. Our study's preliminary evidence of its utility for increasing detection and reporting of probable EA cases offers hope of a true advancement in the field of geriatric medicine and brings us a step closer to addressing the public health quandary of elder maltreatment. Next research steps will focus on validation and further refinement of this instrument, as well as identification of reporting barriers faced by medics. This work is important, timely and salient, and offers many important clinical and policy implications. Given that there are more than 800,000 medics providing services in every county in the U.S. (Mears, 2011), a valid and reliable screening tool that is easy for EMS providers to use could dramatically increase sentinel surveillance of EA in a very short period of time.

Acknowledgments

The authors wish to acknowledge MedStar Mobile Healthcare, Texas Adult Protective Services, John Farris, Amanda Robbins, and the Fort Worth Safe Communities Coalition for their continuing support and dedication in this project.

Funding

This project was supported by Award No. 2014-MU-CX-0102, awarded by the National Institute of Justice, Office of Justice Programs, U.S. Department of Justice. The opinions, findings, and conclusions or recommendations expressed in this publication are those of the author(s) and do not necessarily reflect those of the Department of Justice.

Author contribution statement

BC and JMRG conceived of the study, obtained funding for the study, supervised the conduct of the study and data collection. MBC and DL analyzed the data. JW assisted BC with the conceptual framework of the manuscript as well as with refining interpretation of the results. BC, JMRG, JW, KKJ, and RA drafted the manuscript, and all authors contributed substantially to its revision.

Conflicts of interest

The authors have nothing to disclose.

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